README - R/

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Contents

1 Project Documentation (`R/`)

The `R/` directory contains all R scripts used in the end-to-end workflow of this project. Scripts are organized by milestone and task category to ensure clarity, reproducibility, and modularity.

Each subfolder or script corresponds to a specific phase in the data pipeline, from raw data download to modeling and final outputs. All scripts are written in {tidyverse} style, use relative paths via {here}, and follow consistent documentation and naming conventions.

1.1 File Structure Overview

Subfolder / File	Purpose
`01_download/`	Download raw data (e.g., USGS gage data, climate normals, terrain)
`02_clean_validate/`	Validate, filter, and preprocess raw data into modeling-ready formats
`03_covariates/`	Extract covariates
`04_explore_model/`	Exploratory modeling, collinearity checks, and initial model fitting
`05_final_model/`	Final model selection, generation
`06_eval/`	Model evaluation, and prediction surface
`utils/`	Shared utility functions used across scripts
`*_setup.R`	(Optional) Project-wide setup scripts (e.g., library loading, options)

1.2 Script Documentation

Each major subfolder includes a README.Rmd and README.md that describe:

- Input and output files
- Dependencies
- Workflow summary
- Script-specific notes

Each script begins with a standardized header block including:

- Script name, author, date created and last updated, changelog
- Purpose and generalized workflow
- Inputs and outputs
- Required packages
- Related Files

1.3 Conventions

- File names use a numeric prefix (e.g., 01a_download_usgs_gages.R) for ordering
- Use of here::here() for all file paths ensures portability
- Outputs are written to data/processed/ or data/derived/ with clear filenames
- Scripts are written to be reproducible and modular (no hard-coded paths or state)

1.4 Reproducibility Notes

- Ensure the project environment is managed via {renv}
- Scripts are intended to run independently if inputs are present